Does the Distance Approach to Pairs Trading Survive in the Modern Era?

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Abstract

This study tests whether the distance approach to pairs trading remains profitable for US large-cap stocks over the last decade, from June 2014 to May 2024. Pairs are formed within sectors under the Euclidean, Manhattan and Chebyshev distances, ranked either by the distance alone or by a composite score that also rewards zero crossings and spread volatility. The distance approach does not survive in the modern era. No portfolio earns a significant return, with annualized means between −0.79% and 0.44%, and neither the composite score nor the choice of distance makes a significant difference. The COVID half-year from December 2019 to May 2020 is the best period for every portfolio, and outside it all six portfolios lose money.[1]

1 Introduction

A pair trade holds offsetting positions in two assets whose prices share a long-run equilibrium, and profits when a temporary deviation from that equilibrium closes. The simplest way to identify pairs is the distance approach of Gatev et al. (2006). It matches stocks whose normalized price paths lie closest together over a formation year and trades the pair whenever the two paths diverge by more than two historical standard deviations. Applied to all liquid US stocks from 1962 to 2002, the rule earned excess returns of up to 11% a year.

Whether the rule still works is less clear. Do and Faff (2010, 2012) extended the sample to 2009 and found that profitability declined, mainly because a growing share of pairs never converged, and that the baseline rule became largely unprofitable once trading costs were charged. They improved it by matching stocks only within industries and by favoring pairs with many zero crossings of the spread in the formation period. Krauss (2017) further argued that ranking by the sum of squared distances is analytically suboptimal, since it selects low-variance spreads and therefore trades that earn little.

This study asks whether the distance approach survives in the modern era, using US large-cap stocks at daily frequency from June 2014 to May 2024. We also examine whether a composite score that adds zero crossings and spread volatility to the distance improves performance. We find that the distance approach does not survive. No portfolio earns a mean return distinguishable from zero, the composite score and the choice among Euclidean, Manhattan and Chebyshev distances make no significant difference, and all six portfolios lose money outside the COVID half-year.

The remainder of the report is organized as follows. Section 2 describes the data. Section 3 states the formation rule, the trading rule and the hypotheses. Section 4 reports the results, and Section 5 concludes.

2 Data

The universe comprises a list of 189 large-cap US stocks compiled in 2024. Daily adjusted closing prices from June 2013 to May 2024 are collected, and Table 1 reports the number of stocks and within-sector pairs in each sector.

[1] Code and replication data are available at *GitHub*.

Sector	Stocks	Pairs
Financial Services	24	276
Industrials	21	210
Healthcare	19	171
Consumer Cyclical	19	171
Utilities	18	153
Real Estate	17	136
Consumer Defensive	17	136
Technology	15	105
Communication Services	15	105
Basic Materials	12	66
Energy	12	66
Total	189	1,595

Table 1: Stocks and within-sector pairs by sector.

3 Methodology

The sample is divided into 20 consecutive trading periods E_k of six months, $k = 1, \ldots, 20$, the first beginning in June 2014 and the last ending in May 2024. Each trading period is preceded by a 12-month formation period Φ_k.

Write $P_{i,t}$ as the adjusted closing price of stock i on day t. Within a window $W \in \{\Phi_k, E_k\}$ beginning on day τ_W, normalized prices and pair spreads are defined as

$$\tilde{P}_{i,t} = \frac{P_{i,t}}{P_{i,\tau_W}}, \qquad d_{ij,t} = \tilde{P}_{i,t} - \tilde{P}_{j,t}. \tag{1}$$

3.1 Pair Formation

A pair is eligible in period k when both stocks belong to the same sector and have a complete price record over Φ_k. Over the formation period, let σ_{ij} be the standard deviation of the spread and Z_{ij} the number of zero crossings, defined as the number of sign changes of $d_{ij,t}$. The Euclidean, Manhattan and Chebyshev distances are

$$D_{ij}^{(2)} = \Big(\sum_{t \in \Phi_k} d_{ij,t}^2 \Big)^{1/2}, \qquad D_{ij}^{(1)} = \sum_{t \in \Phi_k} |d_{ij,t}|, \qquad D_{ij}^{(\infty)} = \max_{t \in \Phi_k} |d_{ij,t}|, \tag{2}$$

respectively. The Euclidean distance ranks pairs exactly as the sum of squared distances of Gatev et al. (2006). With T the number of formation days and $\bar{d}_{ij}$ the mean spread,

$$\frac{1}{T} \sum_{t \in \Phi_k} d_{ij,t}^2 = \frac{T-1}{T} \sigma_{ij}^2 + \bar{d}_{ij}^2, \tag{3}$$

so a low distance implies a low spread volatility, as argued in Krauss (2017). The composite score therefore rewards volatility and zero crossings alongside a small distance,

$$R_{ij}^{(q)} = 0.6\, \phi\big(1/D_{ij}^{(q)}\big) + 0.2\, \phi\big(Z_{ij}\big) + 0.2\, \phi\big(\sigma_{ij}\big), \qquad q \in \{2, 1, \infty\}, \tag{4}$$

where ϕ scales a statistic linearly onto $[0, 1]$ over all eligible pairs. For each distance q and each selection $m \in \{\text{distance}, \text{composite}\}$, the portfolio holds the 20 pairs with the lowest $D_{ij}^{(q)}$ or the highest $R_{ij}^{(q)}$, which gives six portfolios.

3.2 Trading Strategy

Prices are normalized again at the start of each trading period, and the position $n_{ij,t} \in \{-1, 0, 1\}$ set at the close of day t follows the rule of Gatev et al. (2006),

$$n_{ij,t} = \begin{cases} -\operatorname{sgn}(d_{ij,t}) & \text{if } n_{ij,t-1} = 0 \text{ and } |d_{ij,t}| > 2\sigma_{ij}, \text{ and } t \text{ is not the last day of } E_k, \\ 0 & \text{if } n_{ij,t-1} \neq 0 \text{ and } d_{ij,t}\, n_{ij,t-1} \geq 0, \text{ or } t \text{ is the last day of } E_k, \\ n_{ij,t-1} & \text{otherwise.} \end{cases} \tag{5}$$

A trade opened at the close of day o buys one dollar of the cheaper stock and sells one dollar of the other, then holds both until the spread crosses zero or the period ends. The portfolio return r_t is the average over its 20 pairs,

$$r_t = \frac{1}{20} \sum_{(i,j)} \left[n_{ij,t-1} \left(\frac{P_{i,t} - P_{i,t-1}}{P_{i,o}} - \frac{P_{j,t} - P_{j,t-1}}{P_{j,o}} \right) - 2c\,|n_{ij,t} - n_{ij,t-1}| - \frac{b}{252} \frac{P_{s,t-1}}{P_{s,o}} |n_{ij,t-1}| \right], \tag{6}$$

where s is the stock held short. The transaction cost is $c = 10$ basis points per dollar traded on each leg, so a round trip costs 40 basis points, and the borrowing fee is $b = 25$ basis points a year on the value of the short leg, following D'Avolio (2002).

Write $r_t^{(q,m)}$ for the daily return of the portfolio with distance q and selection m. We test the three hypotheses

$$\begin{aligned} &\text{H1:}\ \ \mathbb{E}\big[r_t^{(q,m)}\big] > 0, \\ &\text{H2:}\ \ \mathbb{E}\big[r_t^{(q,\text{composite})} - r_t^{(q,\text{distance})}\big] \neq 0, \\ &\text{H3:}\ \ \mathbb{E}\big[r_t^{(q,m)} - r_t^{(2,m)}\big] \neq 0, \end{aligned} \tag{7}$$

where H1 asks whether each portfolio survives, H2 whether the composite score differs from the distance, and H3, for $q \in \{1, \infty\}$, whether the choice of distance matters. We test each with the Newey and West (1987) t statistic for the mean daily return or difference, one-sided for H1 and two-sided for H2 and H3.

4 Results

Table 2 shows that distance-based pair formation does not survive in the modern era. With an annualized volatility of about 4%, no portfolio earns a return distinguishable from zero. The Manhattan distance under distance selection earns the highest return, 0.44% a year, with a Sharpe ratio of 0.11.

Distance	Selection	Return (%)	Volatility (%)	Sharpe	t	p
Euclidean	Distance	0.24	3.88	0.063	0.20	0.42
	Composite	0.17	3.98	0.043	0.14	0.45
Manhattan	Distance	0.44	3.91	0.11	0.36	0.36
	Composite	0.17	4.06	0.041	0.13	0.45
Chebyshev	Distance	−0.79	3.91	−0.20	−0.67	0.75
	Composite	−0.26	4.04	−0.065	−0.21	0.58

Table 2: Net performance, June 2014 to May 2024. Return and volatility are annualized, and p is the one-sided p value.

Figure 1 shows that the weakness is persistent rather than a product of the pandemic. After a small gain through mid-2015, every portfolio drifts downward, apart from a rebound in 2018, until late 2019. The COVID trading period from December 2019 to May 2020, shaded in the

figure, first deepens the losses as spreads widen in March 2020 and then recovers them sharply as the spreads close. After that, the portfolios gain until late 2022 and return most or all of that gain by 2024.

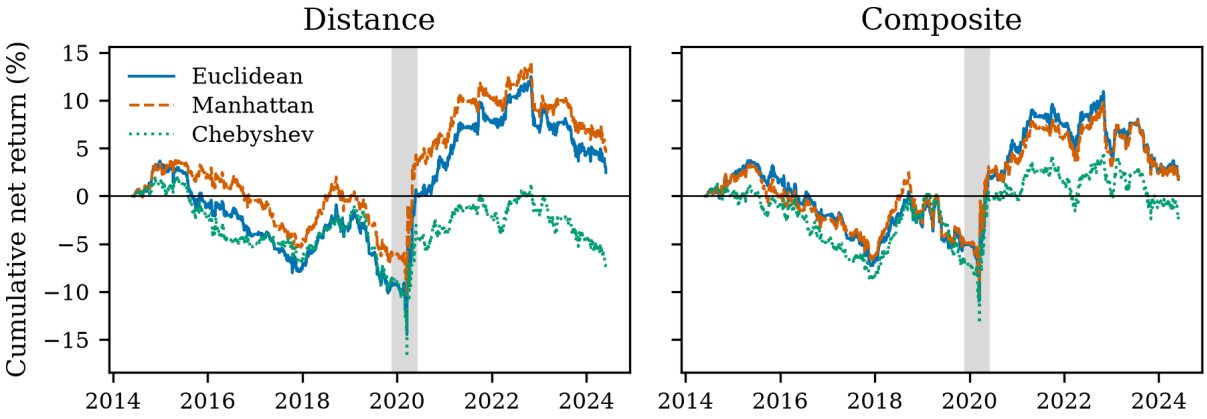


Figure 1: Cumulative net return. The shaded area marks the COVID trading period from December 2019 to May 2020.

Table 3 makes the same point in numbers. Before the pandemic, all six portfolios lose money. The COVID half-year is the best period in the sample for every portfolio, although its returns are not significant. Afterward, the mean returns are close to zero. With the COVID half-year removed, all six portfolios lose money over the remaining 19 periods. The pandemic was therefore not the cause of the poor performance but the one episode that masked it.

		Before COVID		COVID		After COVID		Excluding COVID	
Distance	Selection	Return	t	Return	t	Return	t	Return	t
Euclidean	Distance	−1.67	−1.41	18.9	1.07	0.58	0.33	−0.72	−0.71
	Composite	−0.92	−0.67	14.3	0.90	−0.056	−0.030	−0.56	−0.50
Manhattan	Distance	−1.10	−0.94	20.0	1.11	0.17	0.094	−0.57	−0.56
	Composite	−0.90	−0.64	15.2	0.93	−0.23	−0.12	−0.61	−0.54
Chebyshev	Distance	−1.59	−1.30	8.77	0.48	−0.85	−0.57	−1.28	−1.35
	Composite	−1.25	−0.96	13.1	0.74	−0.55	−0.32	−0.95	−0.91

Table 3: Annualized net mean return (%) and t statistic from June 2014 to November 2019, in the COVID trading period from December 2019 to May 2020, from June 2020 to May 2024 and over all periods except the COVID period.

Neither refinement of the formation rule changes this conclusion. The upper panel of Table 4 compares the composite score with the distance it builds on. The difference is small, changes sign across distances but is never significant. The lower panel compares each distance with the Euclidean benchmark. The Manhattan distance selects nearly the same pairs and earns nearly the same return. The Chebyshev distance, which judges a pair by its single largest divergence, earns about 1% a year less under the distance selection, but the difference falls short of significance with $p = 0.11$.

Table 5 explains why rewarding spread volatility fails to raise returns. The composite score selects pairs with 4% to 7% higher spread volatility and three to five more zero crossings, yet it opens barely more trades and closes a similar share of them by convergence. Converged trades earn slightly more under the composite score, but trades forced to close at the end of the period also lose slightly more. Hence, the profit per trade falls under the Euclidean and Manhattan distances and rises only under the Chebyshev distance, in line with the insignificant differences of Table 4. Across all portfolios, only 36% to 41% of trades converge within their six-month

		Difference (%)	t	p
H2: composite minus distance selection				
Euclidean		−0.072	−0.12	0.90
Manhattan		−0.28	−0.40	0.69
Chebyshev		0.53	0.90	0.37
H3: Manhattan or Chebyshev minus Euclidean				
Manhattan	Distance	0.20	0.50	0.62
	Composite	−0.0059	−0.015	0.99
Chebyshev	Distance	−1.03	−1.60	0.11
	Composite	−0.43	−0.63	0.53

Table 4: Differences in annualized net mean return, with two-sided p values.

window, and the gains of about 6% on those trades are offset by losses of about 4% on the rest. This non-convergence, which Do and Faff (2010) identified as the main source of declining profitability, persists in the modern sample.

						Net profit per trade (%)		
Distance	Selection	σ (%)	Z	Trades/pair	Converged (%)	All	Converged	Forced
Euclidean	Distance	2.55	21.5	1.23	38.6	0.10	6.32	−3.81
	Composite	2.70	25.1	1.26	40.5	0.068	6.69	−4.44
Manhattan	Distance	2.59	22.3	1.23	39.6	0.18	6.43	−3.93
	Composite	2.78	25.8	1.26	39.5	0.066	6.92	−4.41
Chebyshev	Distance	2.58	18.5	1.16	36.1	−0.34	5.80	−3.81
	Composite	2.69	23.0	1.20	37.2	−0.11	6.30	−3.91

Table 5: Formation statistics of the selected pairs and trades. Z is the number of zero crossings. Trades per pair counts trades per selected pair and period. A trade is converged when the spread crosses zero and forced when it is closed at the end of the trading period.

5 Conclusion

In this study, we tested whether distance-based pair formation survives in the modern era. We formed pairs among 189 US large-cap stocks within sectors under three distances (Euclidean, Manhattan and Chebyshev), and ranked them by the distance alone or by a composite score that adds zero crossings and spread volatility.

Backtesting the rule of Gatev et al. (2006) over 20 six-month periods from June 2014 to May 2024, we found that it does not survive. No portfolio earns a mean return distinguishable from zero, and neither the composite score nor the choice of distance improves the result. The COVID half-year is the best period for every portfolio, and all six portfolios lose money without it. Rewarding spread volatility enlarges the gains of converging trades and the losses of non-converging trades alike, so it does not overcome the non-convergence that erodes the strategy.

It should be noted that the universe was compiled in 2024, contains only surviving firms and carries its 2024 sector labels, and that trades are executed at the close that generates the signal. The survivor universe and the same-close execution favor the strategy, so the negative conclusion is, if anything, conservative. Appendix A confirms that waiting one day before trading further lowers every portfolio.

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A Execution with a One-Day Wait

Executing a trade at the close that generates its signal may overstate returns through bid-ask bounce, which Gatev et al. (2006) address by waiting one day. Table 6 repeats the analysis with every position change in (5) executed at the close of the following day. Every portfolio earns less than in Table 2, no mean is distinguishable from zero and the composite score again makes no significant difference, with $|t|$ at most 0.93 for H2.

Distance	Selection	Return (%)	Sharpe	t
Euclidean	Distance	0.073	0.019	0.060
	Composite	−0.062	−0.016	−0.051
Manhattan	Distance	0.36	0.097	0.30
	Composite	−0.13	−0.032	−0.10
Chebyshev	Distance	−1.02	−0.27	−0.91
	Composite	−0.50	−0.13	−0.42

Table 6: Net performance with execution one day after the signal, June 2014 to May 2024.